

AJCR Mathematical Status for Human Review

Human mathematician handoff

24 August 2026

Review state: 24 August 2026.

This handoff is deliberately mathematical. It compares the current route with the supplied execution plan and supervision note; file pointers identify the definitions and theorem statements where each issue lives.

MainProjects/Algebraic-Jacobian-Challenge/informal/Lean_Algebraic_Jacobian_Complete_Execution_Plan.pdf

MainProjects/Algebraic-Jacobian-Challenge/informal/AJCR_Runs_121_122_123_Supervision_Note_2026-08-07.pdf

The mathematical route

The intended curve-specialized route is:

rank-one Picard/divisor chart

- > family-level Abel isomorphism
- > translated rank-one charts over a separably closed field
- > Pic^0 represented over that field
- > finite-Galois descent
- > Pic^0 represented over the original field
- > Jacobian

Established mathematics

Stage	Status	Mathematical content and pointer
Rank-one chart	Established	The good locus of line bundles and the matching divisor locus are defined for arbitrary test schemes. Pic0CriticalPath.lean:205
Family Abel map	Established	The evaluation divisor gives a natural two-sided inverse, rather than only fieldwise uniqueness. Pic0CriticalPath.lean:618
Sep-closed cover	Established	Translations of the rank-one chart cover Picard classes after passing to a separably closed field. Pic0CriticalPath.lean:640
Sep-closed Pic^0	Established	A scheme and a representing universal class exist over the separably closed field; the same pair feeds the current Picard/Jacobian datum. Pic0SepClosedRepresentable.lean:443

The rank-one family producer emphasized in the review PDFs is therefore no longer the mathematical obstruction.

Current descent strategy

The current strategy starts with the separably-closed representing scheme and attempts to spread it to a finite field of definition:

finite affine charts and overlap maps for the separably-closed representer
 -> finite-stage glued scheme `P.gluedOver`
 -> one universal Picard class on `P.gluedOver`
 -> natural equivalence: maps to `P.gluedOver` $\leftrightarrow$ `Pic0` families
 -> projectivity or affine-orbit control for that same scheme
 -> finite-Galois quotient and descent to the original field
 -> `PicRepDatum` -> `JacobianData`

The candidate scheme is defined at:

`Pic0FiniteStageGluedOver.lean:1`

The mathematical decomposition is recorded at:

`AJCR.review-plan.p7-galois-descent.yaml`

Open mathematical obligations

Obligation	Status	Mathematical question and pointer
Universal property of the finite carrier	Open theorem	Does <code>P.gluedOver</code> represent the finite-stage <code>Pic0</code> functor? The required result is a natural equivalence between maps into this one scheme and <code>Pic0</code> families on every test scheme. <code>p7-galois-descent.representability.finite-stage.yaml</code>
Coherence of the universal class	Open theorem	Do the local universal classes, restrictions, and overlap identifications assemble into one natural class on <code>P.gluedOver</code> ? <code>Pic0FiniteStageUniversalClass.lean:1</code>
Exact-carrier orbit geometry	Open theorem	Can one prove projectivity, <code>FiniteInAffine</code> , or an equivalent affine-orbit statement for the same carrier <code>P.gluedOver</code> ? <code>p7-galois-descent.orbit-affine.exact-carrier.yaml</code>
Effective Galois descent	Open theorem	Given the finite-stage representer and its semilinear action, can the quotient and the descended universal class be constructed with the required cocycle identities? <code>DivisorScheme.tex:1777</code>
Original-field Jacobian datum	Downstream consequence	Does the descended scheme and universal class pass unchanged through <code>PicRepDatum</code> , <code>JacobianData</code> , and the final Jacobian statement? <code>Pic0FiniteGaloisJacobianData.lean:1</code>

The relevant mathematical files are:

`Pic0FiniteStageGluePackage.lean`
`Pic0FiniteStageGluingDiagramIso.lean`
`Pic0FiniteStageGluingOverlapIsoPreSnd.lean`
`Pic0FiniteStageGluingOverlapIsoSnd.lean`
`Pic0FiniteStageGluedComparison.lean`

The blueprint states the representation-preserving descent theorem at:

`DivisorScheme.tex:1846`

Mathematical diagnosis

1. The separably-closed representer is available, but the passage from its finite affine atlas to one universal object is not yet expressed as a single theorem.
2. The current work strengthens local restriction, overlap, and base-change identities. Those identities are necessary, but they do not by themselves prove that the glued carrier represents the functor.
3. The quotient step needs projectivity or affine-orbit control for the exact carrier that represents the functor. A theorem about another carrier cannot discharge this obligation.
4. Until the universal property and exact-carrier geometry are settled, the original-field representer and the Jacobian datum cannot be assembled from one representing object and one universal class.

The route decision

The central mathematical question is:

Is $\mathbf{P.gluedOver}$ genuinely the finite-level representing scheme required by the reviewed descent theorem, or is the current strategy constructing an atlas without yet identifying its universal property?

If it is the correct carrier, the next meaningful milestone is one theorem saying that $\mathbf{P.gluedOver}$ represents the finite-stage $\mathbf{Pic0}$ functor, including its universal class, naturality for every test scheme, and the compatibility needed for Galois descent. If that theorem cannot be stated without changing carriers or adding unrelated hypotheses, the route needs redesign.

BOTTOM LINE — The rank-one and separably-closed mathematics is in place. The unresolved mathematics is concentrated in the finite-stage universal property, the geometry of the exact glued carrier, effective descent of the same representing datum, and the final Jacobian handoff.